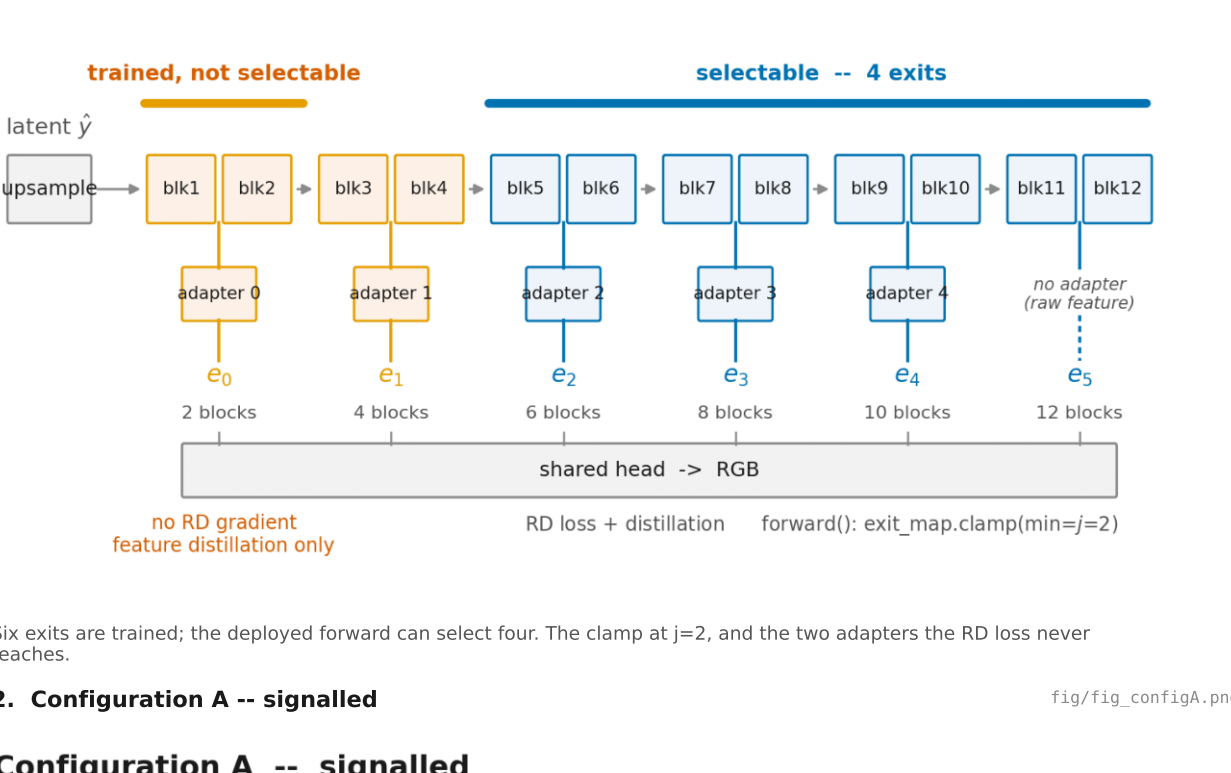


FLEX-UF -- every figure, in the order the argument runs

1. The ladder the main experiment trains

fig/fig1_training.png



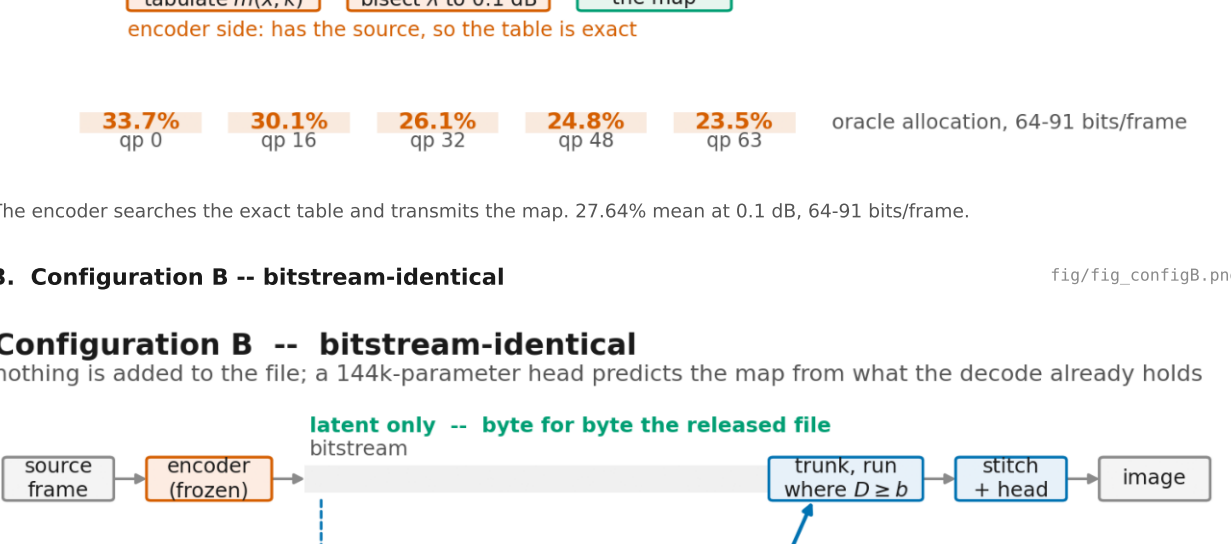
Six exits are trained; the deployed forward can select four. The clamp at $j=2$, and the two adapters the RD loss never reaches.

2. Configuration A -- signalled

fig/fig_configA.png

Configuration A -- signalled

the encoder searches the exact per-cell table and transmits the map; the decoder obeys it



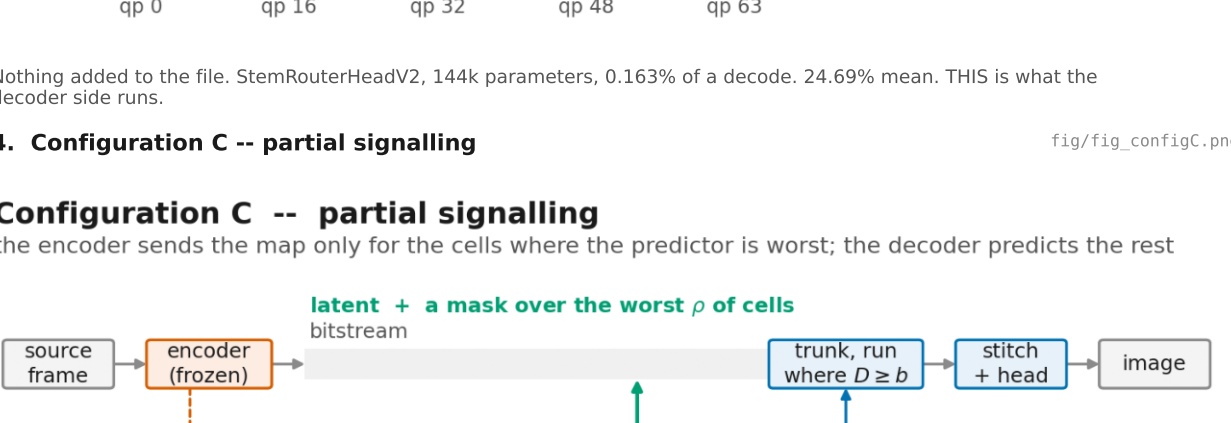
The encoder searches the exact table and transmits the map. 27.64% mean at 0.1 dB, 64-91 bits/frame.

3. Configuration B -- bitstream-identical

fig/fig_configB.png

Configuration B -- bitstream-identical

nothing is added to the file; a 144k-parameter head predicts the map from what the decode already holds



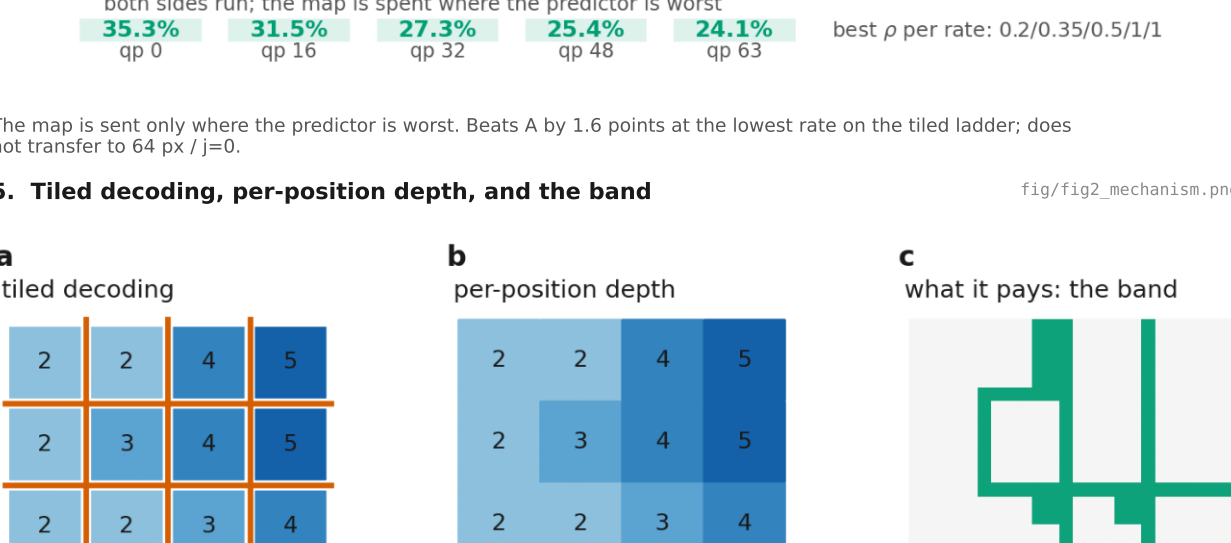
Nothing added to the file. StemRouterHeadV2, 144k parameters, 0.163% of a decode. 24.69% mean. THIS is what the decoder side runs.

4. Configuration C -- partial signalling

fig/fig_configC.png

Configuration C -- partial signalling

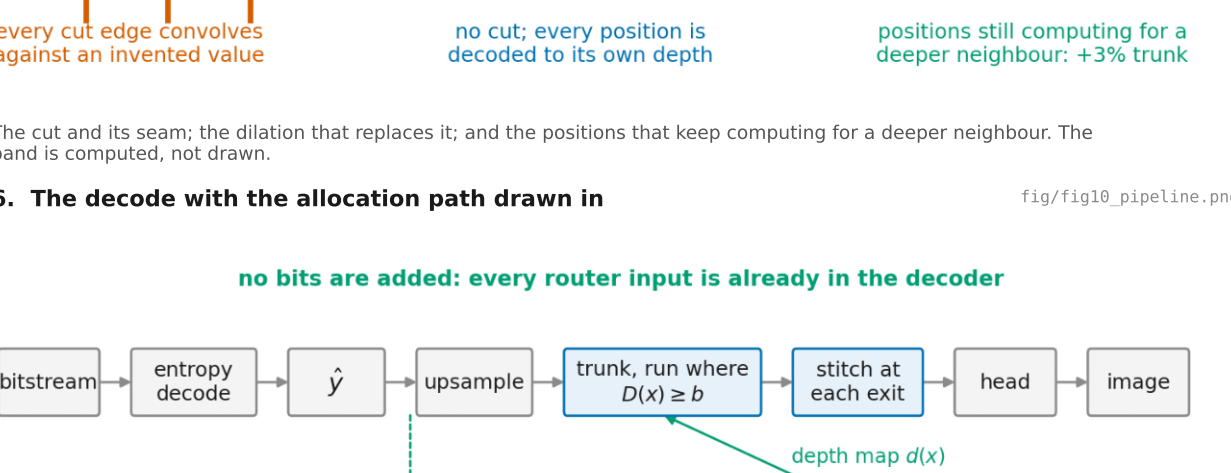
the encoder sends the map only for the cells where the predictor is worst; the decoder predicts the rest



The map is sent only where the predictor is worst. Beats A by 1.6 points at the lowest rate on the tiled ladder; does not transfer to 64 px / $j=0$.

5. Tiled decoding, per-position depth, and the band

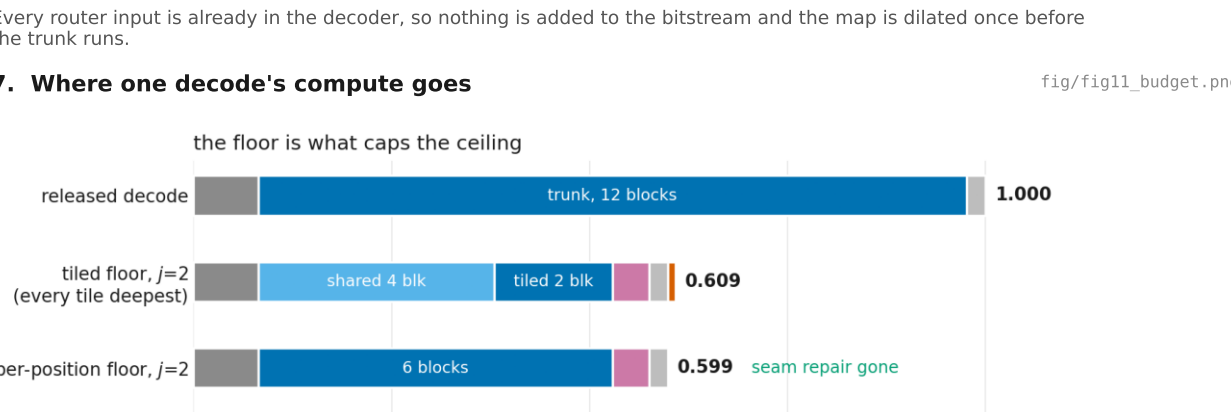
fig/fig2_mechanism.png



The band is cut and its seam; the dilation that replaces it; and the positions that keep computing for a deeper neighbour. The band is computed, not drawn.

6. The decode with the allocation path drawn in

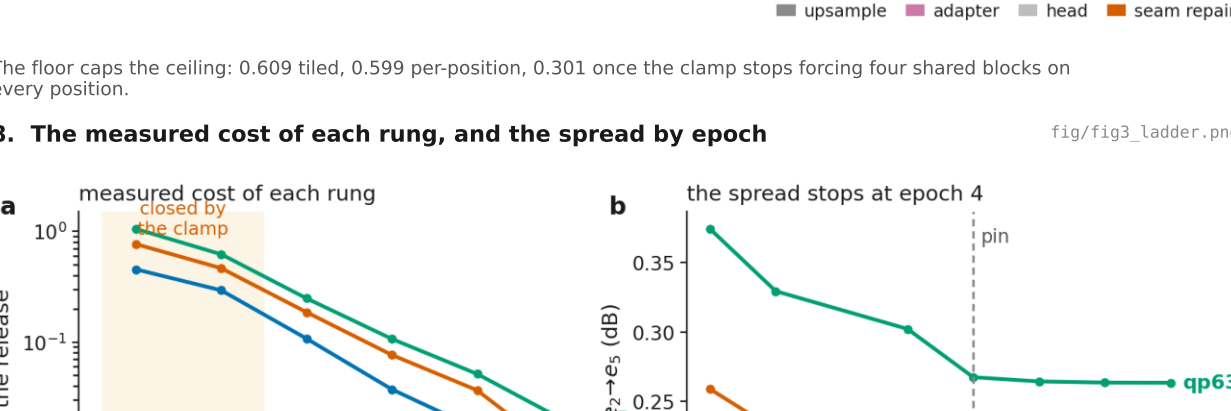
fig/fig0_pipeline.png



Every router input is already in the decoder, so nothing is added to the bitstream and the map is dilated once before the trunk runs.

7. Where one decode's compute goes

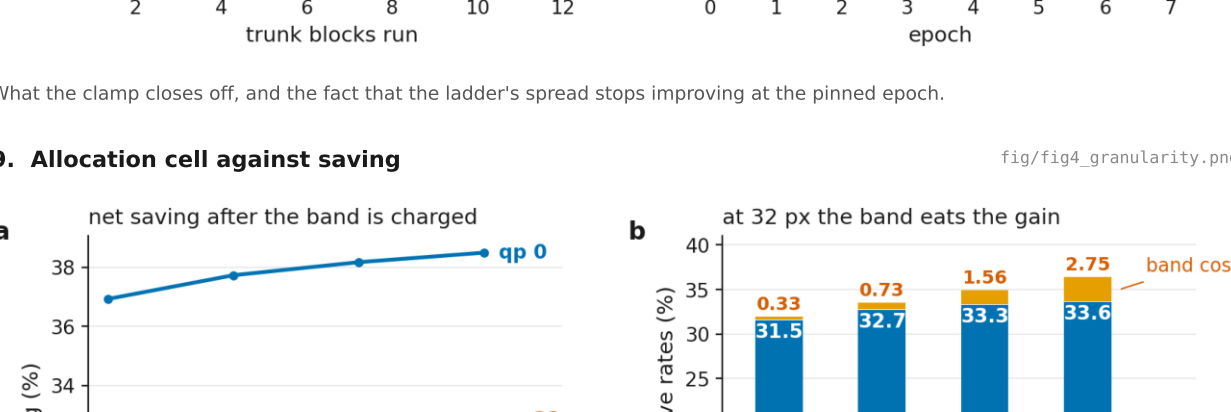
fig/fig11_budget.png



The floor caps the ceiling: 0.609 tiled, 0.599 per-position, 0.301 once the clamp stops forcing four shared blocks on every position.

8. The measured cost of each rung, and the spread by epoch

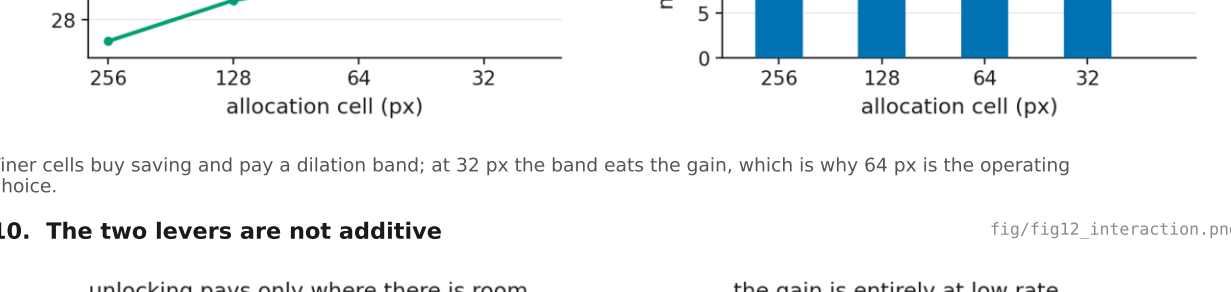
fig/fig3_ladder.png



What the clamp closes off, and the fact that the ladder's spread stops improving at the pinned epoch.

9. Allocation cell against saving

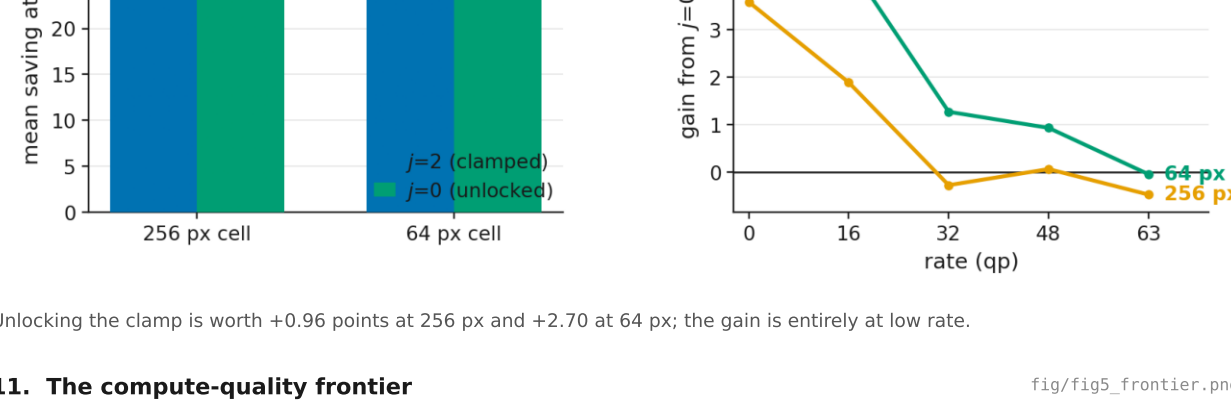
fig/fig4_granularity.png



Finer cells buy saving and pay a dilation band; at 32 px the band eats the gain, which is why 64 px is the operating choice.

10. The two levers are not additive

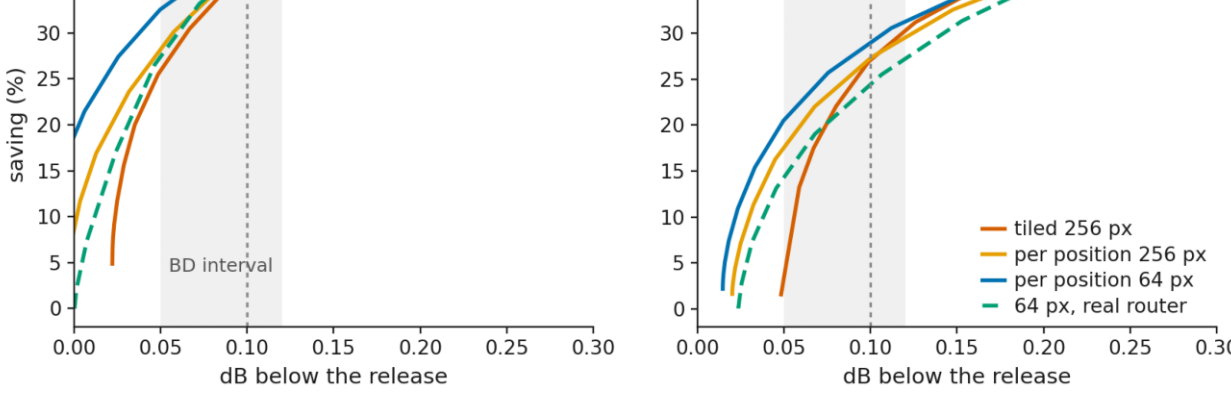
fig/fig12_interaction.png



Unlocking the clamp is worth +0.96 points at 256 px and +2.70 at 64 px; the gain is entirely at low rate.

11. The compute-quality frontier

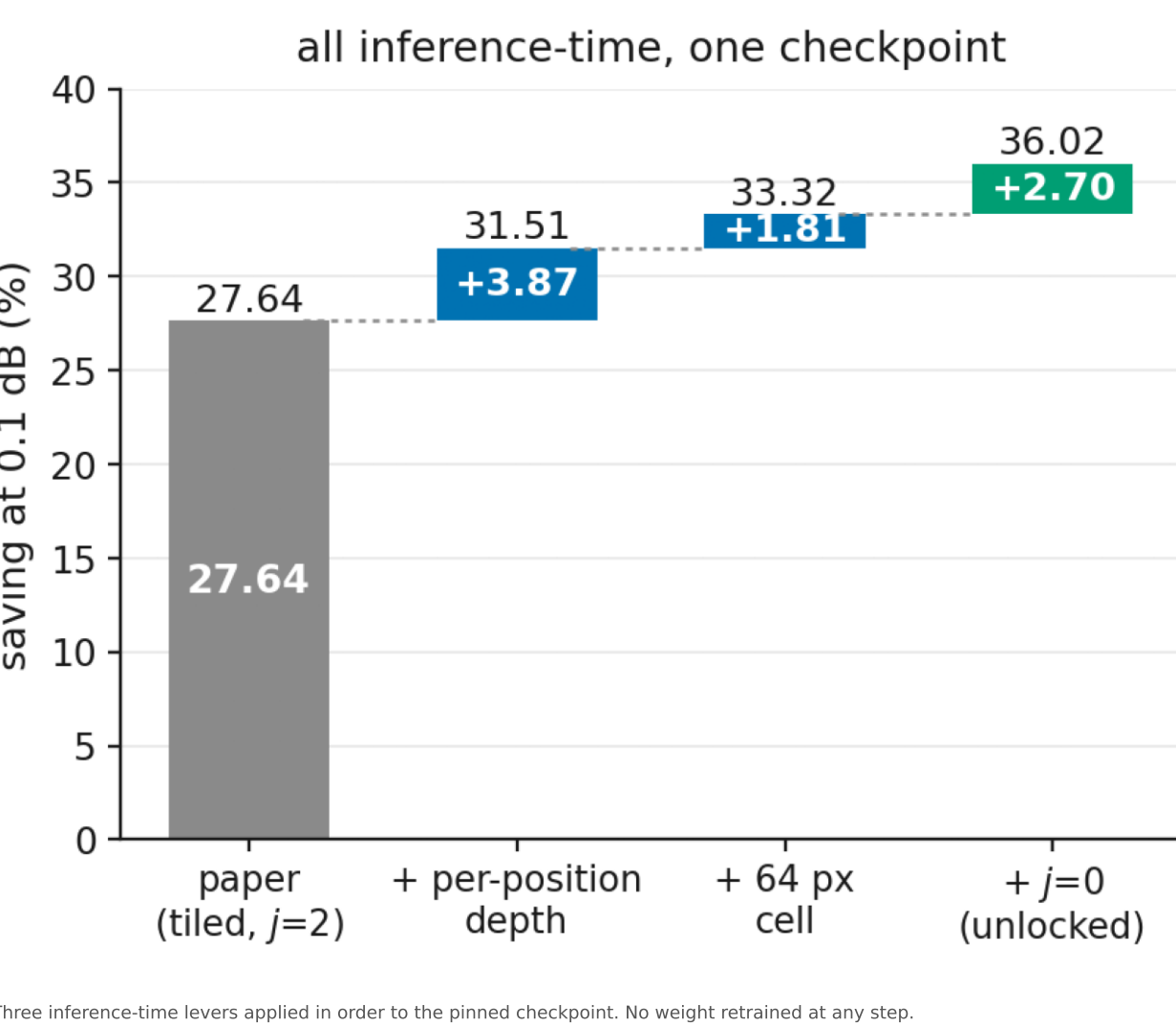
fig/fig5_frontier.png



Tiled against per-position at the extreme rates, with the interval the BD numbers integrate shaded.

12. Where the eight points come from

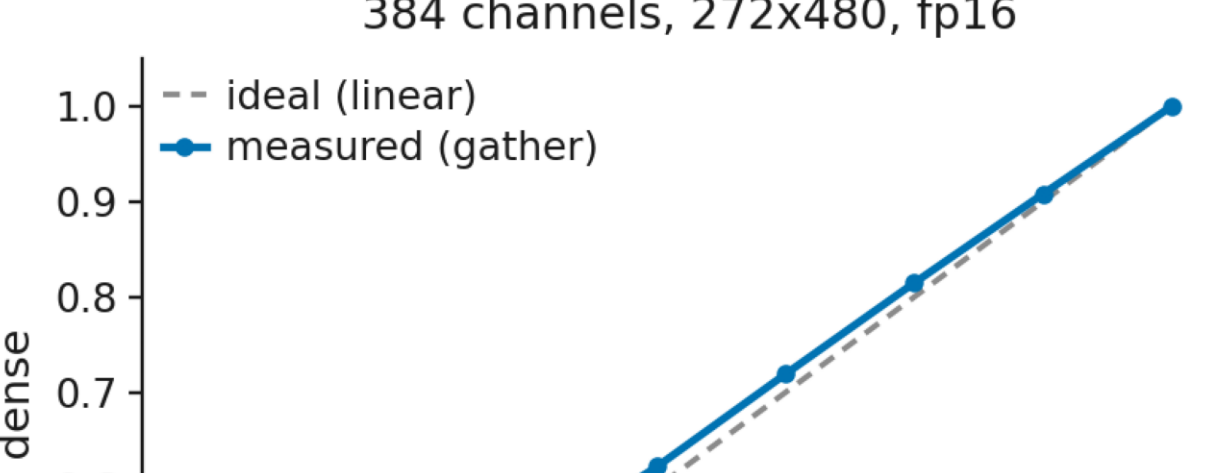
fig/fig6_waterfall.png



Three inference-time levers applied in order to the pinned checkpoint. No weight retrained at any step.

13. Does the time follow the MAC count

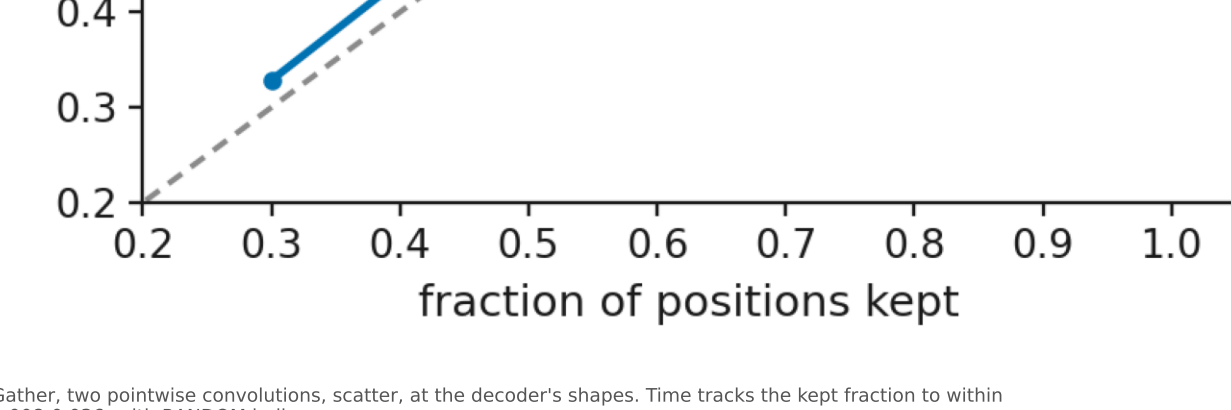
fig/fig7_gather.png



Gather, two pointwise convolutions, scatter, at the decoder's shapes. Time tracks the kept fraction with within 0.008-0.036, with RANDOM indices.

14. The routing axis

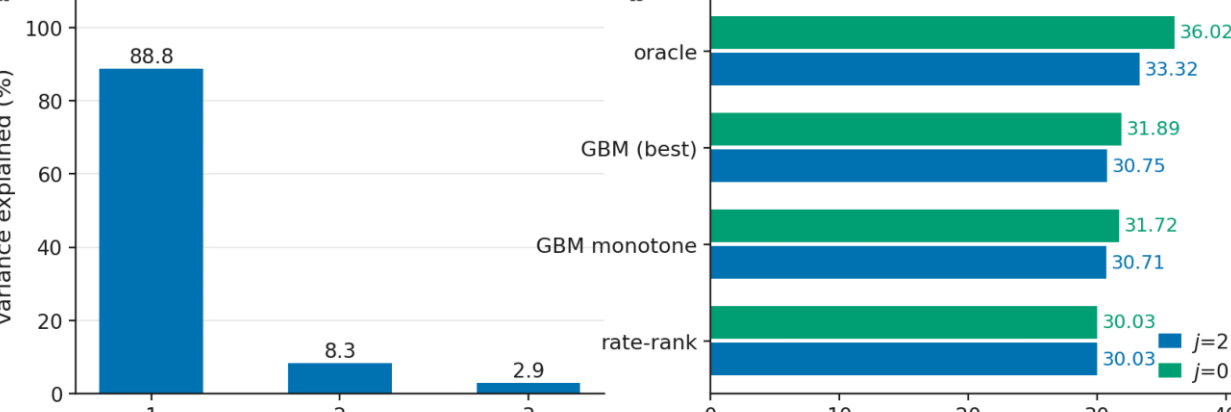
fig/fig9_router.png



The error curve is nearly rank-1 at $j=2$, and the gap between oracle and a real router at both split depths.

15. Saving over rate and budget, and where it saturates

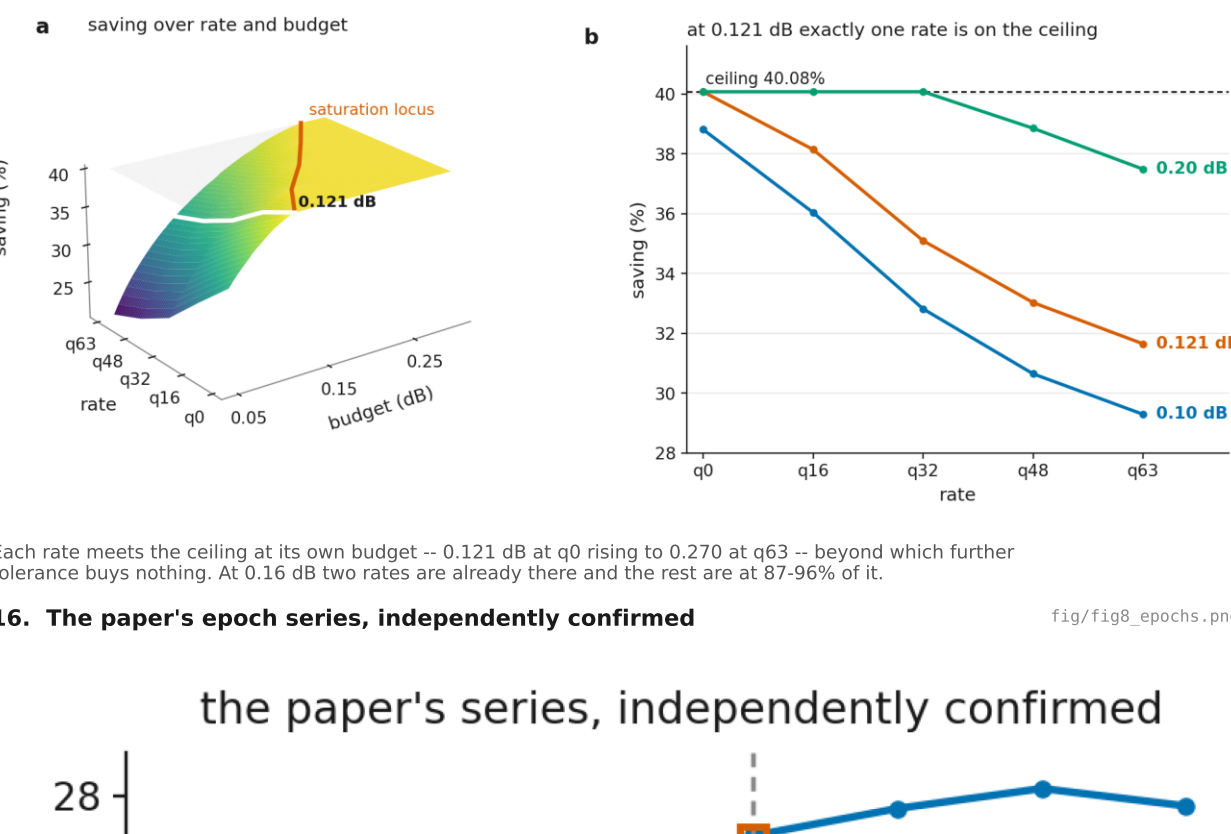
fig/fig13_surface.png



Each rate meets the ceiling at its own budget -- 0.121 dB at q0 rising to 0.270 at q63 -- beyond which further tolerance buys nothing. At 0.16 dB two rates are already there and the rest are at 87-96% of it.

16. The paper's epoch series, independently confirmed

fig/fig8_epochs.png



Re-measured with current scripts on every surviving checkpoint.